kalkulus

latihan soal notasi sigma

1 $\sum_{i=1}^{10} i^2 + \sum_{i=1}^{5} 3i = 385 + 45 = 430$

- $\sum_{i=1}^{10} 3i = 3 \sum_{i=1}^{5} i = 3 \frac{5(5+1)}{2} = 3.15 = 45$

- $\sum_{i=1}^{10} i^2 = \frac{10(10+1)(2(10)+1)}{6} = 385$

2. $\sum_{k=1}^{6} 3k - k^2 = 63 - 91 = -28$

- $\sum_{k=1}^{6} 3k = 3 \cdot \sum_{k=1}^{6} k = 3 \cdot \frac{6(6+1)}{2} = 3.21 = 63$

- $\sum_{k=1}^{6} k^2 = \frac{6(6+1)(2(6)+1)}{6} = 91$

Nama: Muhammad Mahdi Nahassalam

Nim : 245001

3 $\sum_{i=1}^{10} i^2 + \sum_{i=1}^{5} 3i = 385 + 45 = 430$

- $\sum_{i=1}^{10} i^2 = \frac{10(10+1)(2(10)+1)}{6} = 385$

- $\sum_{i=1}^{5} 3i = 3 \cdot \sum_{i=1}^{5} i = 3 \cdot \frac{5(5+1)}{2} = 3 \cdot 15 = 45$

4 $\sum_{j=1}^{4} \sum_{i=1}^{2} a_i\, b_j = (a_1 \cdot b_1) + (a_2 \cdot b_1) + (a_1 \cdot b_2) + (a_2 \cdot b_2) + (a_1 \cdot b_3) + (a_2 \cdot b_3) + (a_1 \cdot b_4) + (a_2 \cdot b_4)$

Latihan Soal notasi Product / Perkalian

1. $\prod_{i=1}^{5} (x_i - 1) = (x_1 - 1) \cdot (x_2 - 1) \cdot (x_4 - 1) \cdot (x_5 - 1)$

2. $\prod_{i=1}^{n} (n-i) = (n-1)(n-2)(n-3)(n-4)(n-5) \ldots\ldots (n-n)$

3 $\prod_{k=4}^{9} k \cdot 2^{k+4} = ((4 \cdot 2)(5 \cdot 2)(6 \cdot 2)(7 \cdot 2)(8 \cdot 2)(9 \cdot 2)^8$

$= ((8)(10)(12)(14)(16)(18))^8$

4. $\prod_{k=0}^{3}(3+k) = (3+0)(3+1)(3+2)(3+3) = (3)(4)(5)(6) = 360$

Latihan Soal binomial newton

1 Jabarkan Setiap Suku Polinomial berikut ini:

a. $(x+2)^3 = x^3 + 3x^2 2 + 3x2^2 + 2^2$
$= x^3 + 6x^2 + 12x + 8$

b. $(2x-3y)^4$
$= \binom{4}{0}(2x)^4(-3y) + \binom{4}{1}(2x)^3(-3y)^1 + \binom{4}{2}(2x)^2(-3y)^2 + \binom{4}{3}(2x)^1(-3y)^3 + \binom{4}{4}(2x)(-3y)^4$
$= 16x^4 - 96x^3y + 216x^2y^2 - 216xy^2 + 81y^4$

2 tentukan koefisien suku x^2y^2 dari Polimonial

a. $(x+y)^4 = \sum_{k=2}^{4}\binom{4}{2}x^{4-2}y^2 = \binom{4}{2}x^2y^2$

$\binom{4}{2} = \frac{4!}{2!(4-2)!} = \frac{4.3}{2.1} = 6$

b. $(3x-2y)^4 = \sum_{k=2}^{4} \binom{4}{2} (3x)^{4-2} (-2y)^2$

$= \binom{4}{2} (3x)^2 (-2y)^2$

$= 6 \cdot 9x^2 \cdot 4y^2 = 6 \cdot 36x^2y^2 = 216x^2y^2$

jadi koefisien dari suku x^2y^2 adalah 216